Predictive and Exploratory Analysis of Social Attitudes Towards Controversial Issues

Computational Social Science
https://github.com/filipponardi17/Controversial-Attitudes-Analysis

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1 Introduction

In recent years, shifting societal attitudes on controversial topics like abortion, same-sex marriage, and immigration have profoundly influenced public policy, social cohesion, and individual rights. As these issues dominate global discourse, understanding the demographic drivers behind these attitudes becomes increasingly critical.

This research aims to answer the question: "What are the underlying demographic factors that shape societal attitudes towards controversial issues, and can these factors reliably predict such attitudes?" To address this, I will leverage data from the World Values Survey, employ exploratory techniques and machine learning algorithms to uncover patterns and predict public opinion trends.

By examining the intricate relationship between demographic characteristics and attitudes toward controversial issues, this research seeks to provide valuable insights into the dynamics of public opinion on key societal issues.

2 Methodology

2.1 Previous Work

This section reviews recent studies analyzing social attitudes using predictive analytics from demographic and social media data.

Gao et al. (2014) developed a computational model to estimate user attitudes on controversial topics through sentiment and opinion analysis on social media(Gao et al., 2014). Predictive Analytics: A Sociological Perspective (2023) highlights the importance of both technical and societal factors in predicting social attitudes(Egbert, 2023).

A 2022 study on migration used machine learning to detect sentiment and stance on social media regarding migration(Chen et al., 2022). AlDayel and collegues (2021) focused on stance detection, emphasizing the influence of personal, cultural, and social factors on users' opinions about controversial issues (AlDayel and Magdy, 2021).

The study that influenced me the most is Jafarigol et al. (2023), which utilized machine learning to explore global religious affiliations using World Values Survey data. Their research identified key factors influencing religiosity across 30 countries. This approach underscores the relevance of demographic factors in shaping religious beliefs and behaviors (Jafarigol et al., 2023).

2.2 Dataset

For this study, I utilized data from the joint European/World Values Survey (EVS-WVS) 2017-2020, which covers a broad dataset collected between 2017 and 2020 EVS (2017-2020). This comprehensive survey provides insights from 90 countries worldwide, with a total of 153,716 respondents.

Data were collected through using mix methods to ensure flexibility and inclusivity. These included:

- Face-to-face interviews: CAPI (Computer-Assisted) and PAPI (Paper-and-Pencil)
- Telephone interviews: CATI (Computer-Assisted)
- Self-administered questionnaires: CAWI (Web-based) and paper-based formats

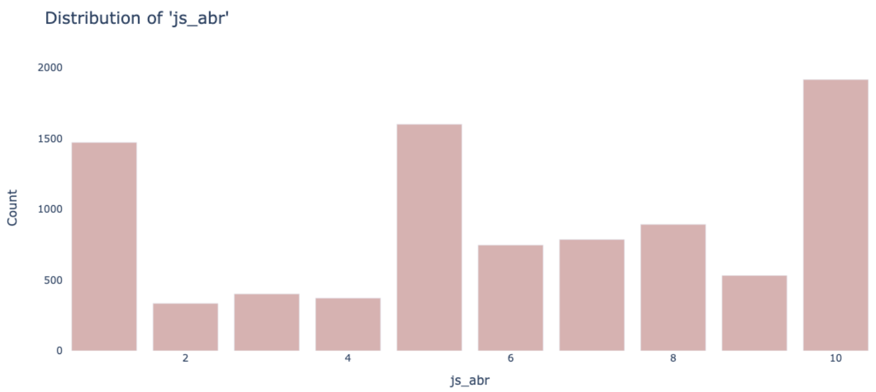
Mixed-mode methods improve research reach and quality by offering various ways to participate, which boosts response rates and reduces costs and interviewer bias. However, these methods also face challenges such as inconsistencies in how questions are perceived across different modes, difficulties in integrating diverse data types, potentially lower quality of data in self-administered surveys, and access barriers to necessary technology for some respondents. My analysis focuses on the four largest European economies by GDP—Germany, France, Spain, and Italy. I selected these countries due to their large respondent base and their relative comparability in terms of economic and social structures.

The codebook for the survey will be in the reference link (World Values Survey Association, 2024), as well as a table to see what the variable name is in the codebook (Filippo Nardi, 2024).

2.3 Data Cleaning and Pre-Processing

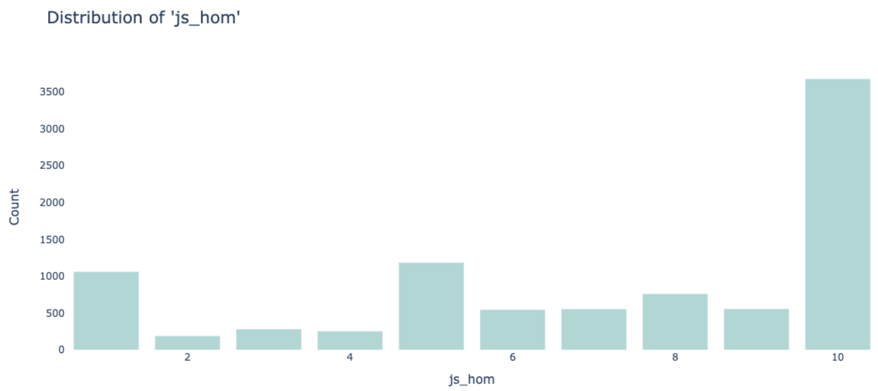
The controversial issues I selected as target variables for this study, and their distribution are as follows:

- **js_abr** which relates to this question in the WVS: Justifiable: Abortion; Please tell me for each of the following whether you think it can always be justified, never be justified, or something in between, using this card. (1 Never justifiable, 10 Always justifiable)



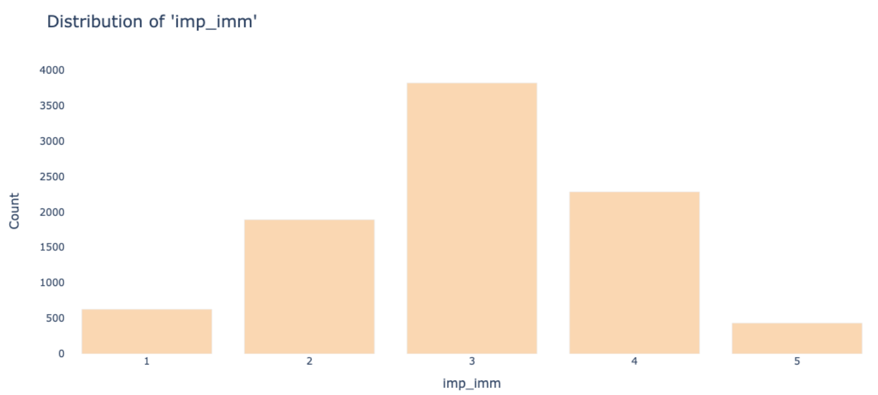
Distribution of **js_abr**

- **js_hom** which relates to this question in the WVS: Justifiable: Homosexuality; Please tell me for each of the following whether you think it can always be justified, never be justified, or something in between, using this card. (1 Never justifiable, 10 Always justifiable)



Distribution of **js_hom**

- **imp_imm** which relates to this question in the WVS: Now we would like to know your opinion about the people from other countries who come to live in [your country] - the immigrants. How would you evaluate the impact of these people on the development of [your country]? (1 Very bad, 5 Very good)

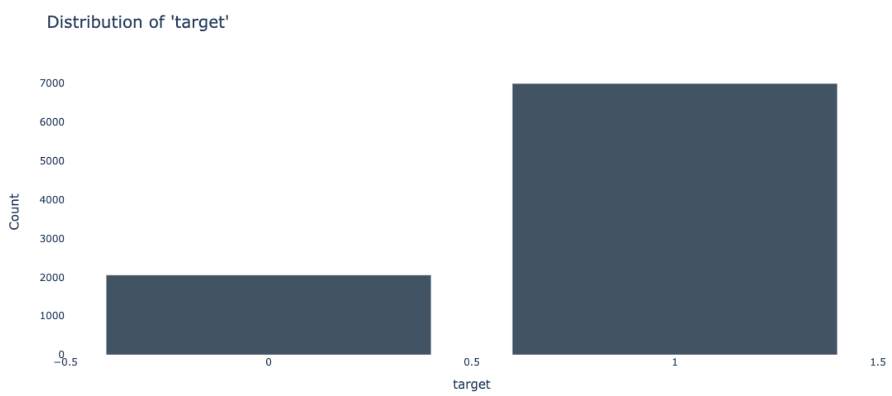


Distribution of **imp_imm**

Missing data posed a challenge for the algorithms I employed, prompting me to consider several imputation techniques. I opted to use PMM due to its simplicity and speed. PMM works by identifying the nearest neighbors of data points with missing values and imputing values from other data points with similar characteristics.

In the data cleaning and preprocessing phase, outliers were removed during the to improve the reliability of the results. I first created individual sub-dataframes for each country. I then combined the data from France, Italy, Germany, and Spain into a single dataframe, referred to as `dfEU`. A new variable, `target`, was created by summing the variables `dfEU['js_abr']`, `dfEU['js_hom']`, and `dfEU['imp_imm']`. To simplify the analysis, this `target` variable was converted into a binary format, with values of either zero or one. Specifically, if the sum of these three variables was less than 12, the target variable was set to zero, indicating a higher prevalence of controversial opinions (as the maximum possible sum was 25). This transformation helped classify individuals based on the predominance of controversial attitudes. This variable was only utilized in the predictive portion of the paper, ensuring that its binary classification served as a key component for the predictive analysis. After this passage, scaling was applied only to ensure the comparability of the variables.

This highlighted a significant imbalance in the distribution of the `dfEU['target']` variable (as detailed in the figure below, with controversial opinions appearing more than three times less frequently than non-controversial ones. This imbalance skewed the predictive models towards non-controversial outcomes.



Distribution of **target**

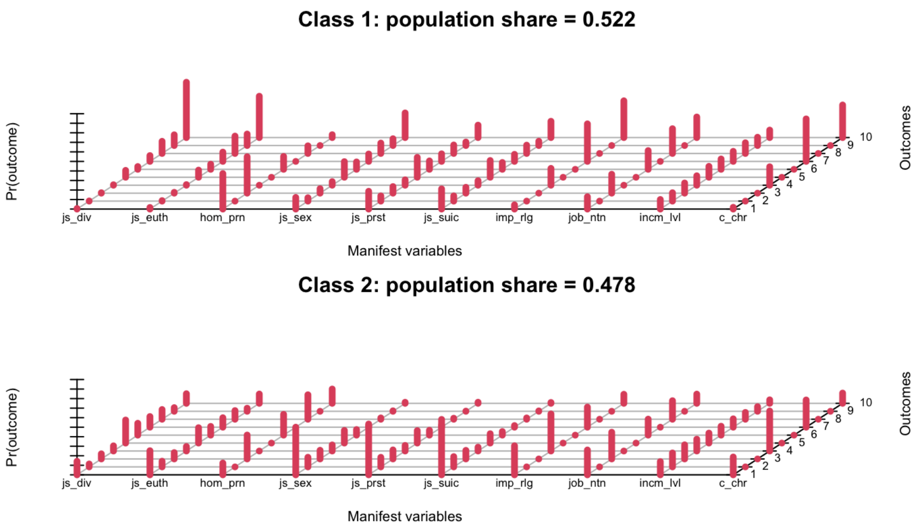
To address this issue, I explored various rebalancing techniques for the training dataset. Interestingly, undersampling proved to be the most effective approach and thus I opted for that one.

3 Exploratory Analysis

In this section, I outlined the Exploratory Analysis conducted using several key techniques. Latent Class Analysis (LCA) was applied to identify hidden groups or patterns within the data, while Principal Component Analysis (PCA) was used to examine the loadings of the first two principal components, capturing the majority of the variance and revealing the dataset's underlying structure. Random Forest Feature Importance further helped evaluate the significance of each feature in predicting the target variable, offering insights into key drivers and guiding feature selection and model refinement. Additionally, Exploratory Factor Analysis (EFA) was performed to uncover latent factors explaining the correlations among variables, providing a deeper understanding of the data structure and enhancing the overall model development.

3.1 Latent Class Analysis

I applied the Latent Class Analysis (LCA) algorithm to gain insights into the subpopulations within my dataset. Given that the target variable was categorized as binary, I opted to use LCA to identify two distinct classes, providing distinction between the primary subgroups in the data. By examining these subpopulations, I could better understand their characteristics and how they contributed to the overall findings of the study. The results of the analysis were as follows:



Latent Class Analysis

The two subpopulations are relatively similar in size, with one comprising 48 percent of the data and the other 52 percent. The variables chosen were a set of distinct social and economic viewpoints. By examining each class in detail, we can better understand their attitudes and values, which provide valuable insights into the broader patterns present in the data. Below is a breakdown of the defining features and ideologies of each class, offering a deeper exploration of how these subpopulations diverge on key social and economic issues.

- **Class 1:** Socially Conservative and Economically Protectionist This group is conservative, with low acceptance of divorce and restrictive views on euthanasia, homosexuality, and prostitution. They oppose suicide and heavily value religion, which shapes their beliefs. Economically, they prefer national over foreign workers and prioritize income equality, showing a protectionist stance. Their confidence in religious institutions is high.

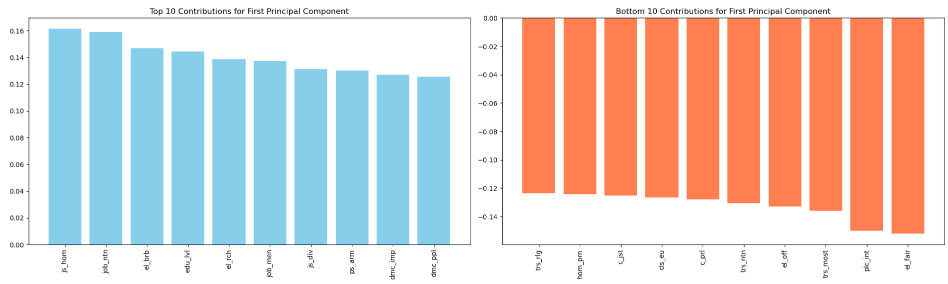
- **Class 2:** Moderately Progressive This group accepts divorce and shows moderate openness towards euthanasia and homosexuality. Their views on sexual norms and prostitution are less conservative, though some reservations persist. They are more permissive about suicide than Class 1. While progressive, they still value religion, indicating an influence of traditional values. Economically, they are less protectionist, more welcoming to foreign workers, and have balanced views on income distribution. Their confidence in religious institutions, while present, is lower than in Class 1.

These results suggest that while both classes adhere to traditional values, they differ in their willingness to embrace social change and their economic priorities. Understanding these distinctions can provide valuable insights for policymakers and researchers, helping to design more effective interventions in public policy and social programs.

3.2 Principal Components Loadings Analysis

To further explore the underlying structure of the dataset, Principal Component Analysis (PCA) was applied. This method reduces the dimensionality of the data while retaining most of its variability, making it an effective tool for identifying key patterns. By examining the loadings of the first two principal components, we gain insights into the most influential variables and their contributions to the overall variance, enhancing our understanding of the dataset's composition.

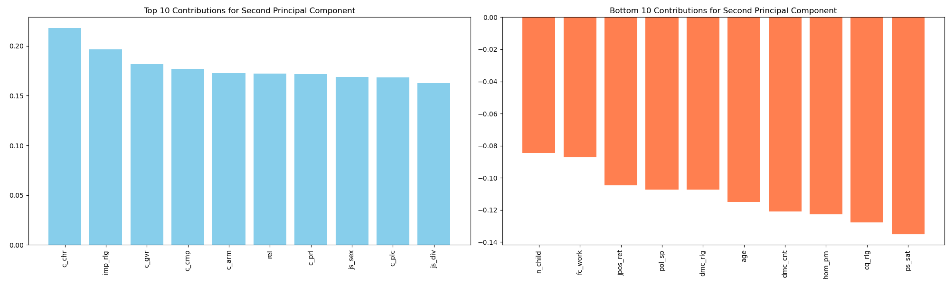
The loadings of the first components reveal significant data structure insights as follows:



Principal Component 1 Loadings

- **Component 1: Perceptions of Justice and Equity:** This factor examines societal norms related to justice, fairness, and equality, focusing on how communities perceive social fairness and treat diverse groups. Key variables like the justifiability of homosexuality (*js_hom*) and job opportunities for different nationalities (*job_ntn*) highlight views on inclusivity and equity. Variables such as the acceptability of divorce (*js_div*) and measures of political representation (*dmc_imp*, *dmc_ppl*) underscore the importance of personal freedoms and fair political voice. Conversely, lower importance variables like trust in people of another religion (*trs_rlg*), confidence in the judiciary (*c_jst*), and electoral integrity (*el_fair*) indicate skepticism about the actual implementation of these justice ideals, pointing to a gap between theoretical values and practical experiences.

Presented below is the plot for the second principal component loadings, highlighting the key variables driving its variance and revealing additional data patterns



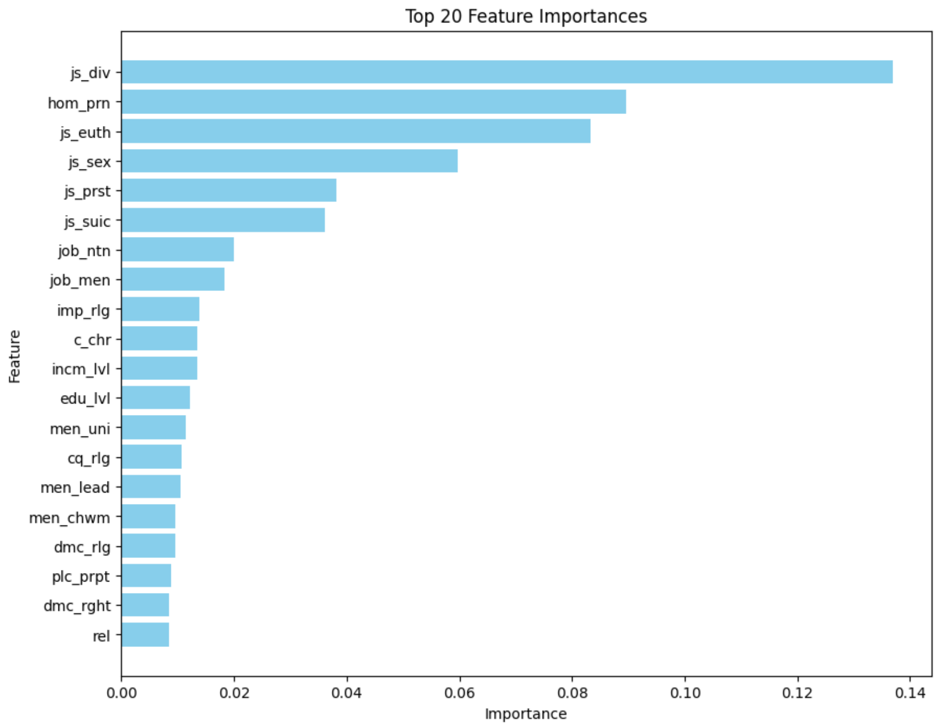
Principal Component 2 Loadings

- **Component 2: Societal Values and Religious Influence:** This factor highlights the strong impact of religious beliefs on societal attitudes and behaviors, with significant variables such as confidence in churches (`c_chr`) and the importance of religion in life (`imp_rlg`) demonstrating this influence. Trust in governmental and social institutions (`c_gvr`, `c_cmp`, `c_prl`) reflects the interplay between religion and institutional confidence. Variables like the justifiability of casual sex (`js_sex`) and divorce (`js_div`) point to the role of religious and societal values in guiding personal and public moral decisions. Conversely, aspects like the number of children (`n_child`), the future importance of work (`fc_work`), and political self-positioning (`pol_sp`) have lower influence from these frameworks. Negative loadings on political system satisfaction (`ps_sat`) indicate a lesser relevance of these areas in the prevailing value systems.

3.3 Random Forest Features Importance

Here, I employed the Random Forest algorithm, using the "target" variable mentioned before. I extracted the 20 most important features based on the Mean Decrease Index, which measures feature importance. This approach helped me identify the top 20 features that have the most significant impact on the predicting controversial issues.

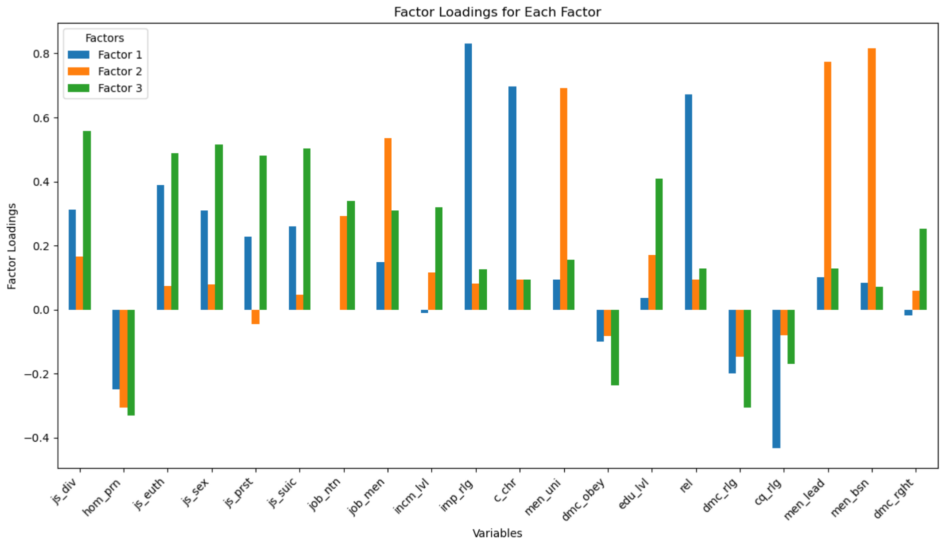
The top 20 variables with their Mean Decrease Index are as follows:



Top 20 Feature Importance

3.4 Factor Analysis

I conducted an exploratory factor analysis (EFA) with the aim of identifying underlying patterns within the data, using only the first 20 variables found in Random Forest Feature Importance. After examining the eigenvalues, I chose to retain three factors, as these had eigenvalues noticeably greater than 1, indicating that they explain a significant proportion of variance in the data compared to other factors. Given that the variables were not independent, I employed an oblimin rotation, which allows for factors to correlate and provides a more realistic representation of complex social data where relationships between variables are expected.



Exploratory Factor Analysis

Here I suggest an explanation of the 3 factors:

- Factor 1: Personal and Ethical Judgments
  This factor had strong positive loadings for variables like js_div (justifiable: divorce), js_euth (justifiable: euthanasia), and js_sex (justifiable: casual sex), indicating its focus on Individual Morality and Personal Judgments. Additionally, edu_lvl (education level) and job_ntn (jobs for nationals) loaded onto this factor, connecting education and employment views to personal ethics.

- Factor 2: Gender Roles and Leadership
  This factor captured attitudes toward traditional gender roles, with strong loadings from men_lead (men better political leaders) and men_bsn (men better business executives). It reflects support for traditional Gender Roles and Leadership.

- Factor 3: Religious and Moral Convictions
  High loadings from imp_rlg (importance of religion) and rel (religious person) show a focus on Religiosity and Spiritual Convictions. Notably, cq_rlg (critical view of religion) had a negative loading, opposing critical religious perspectives.

4 Predictive Analysis

In this section, I conducted predictive analysis using various machine learning algorithms to predict the controversial respondents. These algorithms were selected for their capacity to handle classification tasks and analyze complex relationships within the data. The methods used include:

1. **Random Forest:** An ensemble learning method that constructs multiple decision trees during training. It combines the output of these trees (majority vote) to improve classification accuracy and reduce overfitting.

2. **K-Nearest Neighbors (KNN):** A simple, non-parametric algorithm that classifies a data point based on how closely it resembles the data points in its nearest neighbor set. It predicts the label by majority voting among the k-nearest points.

3. **SVM with Linear Kernel:** This is the simplest form of SVM, where the data is assumed to be linearly separable. It constructs a straight-line decision boundary between classes.

4. **SVM with Radial Basis Function (RBF) Kernel:** The RBF kernel allows the SVM to create a more flexible decision boundary by mapping data points into higher dimensions, making it effective for data that is not linearly separable.

5. **Support Vector Machine (SVM) with Polynomial Kernel:** This SVM variation uses a polynomial function to transform the data, allowing the algorithm to find complex relationships between the input features and the target variable by fitting non-linear decision boundaries.

6. **Logistic Regression:** A generalized linear model used for binary classification. It predicts the probability that a given input belongs to one of two classes by applying a logistic function to a weighted sum of the input features.

7. **Gradient Boosting:** An ensemble technique that builds models sequentially, with each new model correcting the errors of the previous one. It combines the predictions of multiple weak learners (usually decision trees) to create a strong predictive model.

To optimize the performance of each algorithm, I applied a grid search with cross-validation to tune the hyperparameters. This approach systematically evaluates combinations of hyperparameters to identify the best configuration for each algorithm, ensuring that the models are optimized for predictive accuracy before final evaluation.

To assess the performance of each algorithm, I employed the following evaluation metrics:

- **Area Under the ROC Curve (AUC):** AUC quantifies the ability of the model to distinguish between positive and negative classes, where a value closer to 1 indicates better performance.
- **Sensitivity (Recall):** Measures the proportion of actual positives correctly identified by the model, i.e., the true positive rate.
- **Specificity:** Refers to the proportion of actual negatives that are correctly identified, or the true negative rate.
- **Accuracy:** Represents the overall correctness of the model's predictions, calculated as the ratio of correctly predicted instances to the total number of instances.
- **Precision:** Defines the proportion of true positive predictions among all positive predictions, focusing on the accuracy of positive class identification.
- **F1 Score:** A harmonic mean of precision and recall, providing a balanced measure that accounts for both false positives and false negatives. It is particularly useful in cases of imbalanced datasets.

By employing a diverse set of algorithms and evaluation metrics, I aimed to capture a comprehensive understanding of the predictive power of the features related to controversial issues. The use of these various metrics ensures a thorough assessment of each model's performance across different dimensions, enabling a more robust comparison and selection of the most effective model for this task.

5 Results

In this section, I present the results of the predictive analysis, where the machine learning algorithms were evaluated based on their ability to predict the controversial respondents. The table below summarizes the performance of all algorithms, highlighting the predictive power of each approach in capturing the relationships between the demographic features and the target variable for each different performance metric.

Method	AUC	Overall Accuracy	Sensitivity	Specificity	Precision	F1 Score
Random Forest	0.83	0.86	0.89	0.77	0.92	0.91
KNN	0.78	0.83	0.87	0.70	0.90	0.88
Linear SVM	0.82	0.85	0.87	0.77	0.92	0.90
RBF SVM	0.84	0.86	0.88	0.80	0.93	0.91
Polynomial SVM	0.84	0.85	0.86	0.81	0.93	0.90
Logistic Regression	0.84	0.85	0.86	0.82	0.94	0.90
Gradient Boosting	0.84	0.85	0.86	0.81	0.93	0.90

Table 1: Performance metrics of different methods

6 Conclusions

This research successfully identified and analyzed the demographic factors influencing societal attitudes towards controversial topics, specifically abortion, same-sex marriage, and immigration.

Through exploratory analysis, the study uncovered significant divisions between socially conservative and moderately progressive views, emphasizing the roles of justice, fairness, and religious values in shaping public opinions.

The feature importance analysis provided additional insights, identifying key predictors of societal attitudes, such as views on divorce, homosexuality, euthanasia, and other personal freedoms.

The most accurate predictions using machine learning algorithms were achieved with Random Forest and Logistic Regression models, which effectively captured the complex relationships within the data, offering a strong framework for forecasting trends in societal attitudes.

However, the complexity of these methods raises concerns about interpretability and the risk of overfitting, especially with noisy data. While the mixed-mode data collection is comprehensive, it may introduce inconsistencies and limit participation among certain demographics, potentially skewing results. Additionally, cultural and temporal variations may affect the generalizability of the findings, as societal attitudes change rapidly across regions and over time. Therefore, caution is needed when interpreting the predictive models and applying them to different contexts.

Overall, while the research effectively addressed its objectives by accurately predicting societal attitudes and identifying the underlying demographic factors, it also highlights the complexities and challenges inherent in forecasting societal trends.

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